

Tyler Allen

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Senior Backend Engineer specializing in Python APIs and cloud systems with experience building production ML services.

EXPERIENCE

Verifacto

Atlanta, GA

Senior Python Engineer

February 2026 – May 2026

- Designed backend services for AI chatbots with LangChain and AWS Bedrock, and Django to deliver insurance coverage updates to thousands of customers.
- Debugged production issues across distributed PHP, Python, and JavaScript services, cutting weekly incident resolution time in half.

EvidentID

Atlanta, GA

Senior Backend Software Engineer

January 2021 – April 2025

- Built a training framework for a fine-tuned BERT model, including data preparation and evaluation, to automate background check data classification; reduced manual review time from 3 hours to 5 minutes (98% improvement).
- Architected and deployed a production ML classification service on Azure Kubernetes using Python, RabbitMQ for async inference and S3 for model versioning, serving 1000+ predictions/day.
- Modernized the Python stack to improve developer productivity by introducing FastAPI, async libraries, Poetry for dependency management, Docker-based testing, and standardized documentation practices.
- Designed a FastAPI Forward-Auth microservice for Emissary Ingress on Kubernetes, centralizing OAuth/JWT validation; worked with frontend and DevOps teams to standardize authentication across all web applications.
- Developed Scala backend services for insurance document ingestion, processing over 30K policies per month using Akka HTTP, Slick, PostgreSQL, and AWS S3.

Backend Software Engineer

December 2017 – January 2021

- Led an Agile 3-person team to build an automated background check service in Python on AWS EC2, processing 10K+ verifications per month and generating over 40% of company revenue.
- Scaled backend microservices with an event-driven architecture using RabbitMQ, Celery for Python services, and Redis caching achieving a 5x performance improvement in the background check platform.
- Maintained comprehensive end-to-end automated testing with PyTest, Selenium, and Cucumber/Gherkin for Behavior-Driven Development; enabled thorough regression testing for all services.
- Configured Jenkins workflows for CI/CD to run linting, unit and integration tests, builds using Docker and Helm, and pushing to PyPi and Docker repositories; enabled automated deployments for development environments.
- Resolved production incidents during on-call rotations using logs and metrics, improving system reliability with observability tools including Prometheus, Grafana, Logz.io, and Sentry.

Junior Backend Software Engineer

November 2016 – December 2017

- Contributed to the core backend of a secure privacy-focused platform for sharing sensitive data implemented in Scala and using MongoDB.
- Worked within an Agile Scrum team, participating in sprint planning, estimation, code reviews, and retrospectives; led concise daily standups.
- Developed data models and serialization for Evident's Python SDK and built the first data verification services from scratch; deployed on AWS EC2 with Ansible and Terraform.

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science, Focus in AI and Devices

August 2013 – May 2016

SKILLS

Languages: Python, Scala, Elixir

Machine Learning: PyTorch, HuggingFace Transformers, BERT, NLP, LangChain, LLM APIs

DevOps: Kubernetes, Docker, AWS (S3, EC2, IAM), Azure, CI/CD (Jenkins), RabbitMQ

Backend Engineering: PostgreSQL, MongoDB, Redis, SQLAlchemy, REST APIs, FastAPI, Flask

Practices: Test Driven Development, Agile/Scrum, OOP and Functional Programming

Tools: Git, GitHub, Jira, Postman, Swagger, OpenAPI